

Figure 7: Gradle build

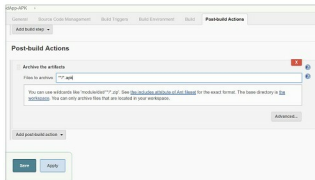


Figure 8: Archive Artifact

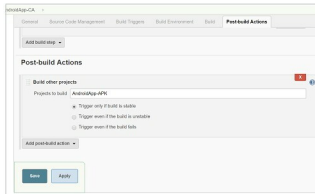


Figure 9: Downstream job

build.gradle file in the Android project.

The Android Lint plugin offers a feature to examine the XML output produced by the Android Lint tool and provides the results on the build page for analysis. It does not run Lint, but Lint results in XML format must be generated and available in the workspace.

Creating a Freestyle project to build the APK file for the Android application: After completing the analysis of the code, the next step is to build the Android project and create an APK file to execute further.

Creating a Freestyle project with the name AndroidApp-APK: In the build actions, select *Invoke Gradle script*.

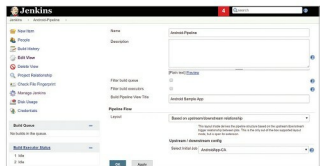


Figure 10: Build Pipeline flow

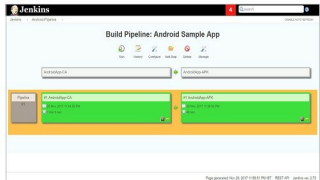


Figure 11: Successful execution of the Build Pipeline

Archive the build artefacts such as JAR, WAR, APK or IPA files so that they can be downloaded later. Click on *Save*.

After executing all the jobs, the pipeline can be pictorially represented by using the Build Pipeline plugin. After installing that plugin, users have to give the start, middle and end points to show the build jobs in that sequence. They can configure upstream and downstream jobs to build the pipeline.

To show all the build jobs, click on the '+' sign on the right hand side top of the screen. Select *build pipeline* view on the screen that comes up after clicking on this sign. Configuring a build pipeline view can be decided on by the user, as per requirements. Select *AndroidApp-CA* as the initial job.

There are multiple options like the Trigger Option, Display Option, Pipeline Flow, etc.

As configured earlier, the pipeline starts by clicking on the *Run* button and is refreshed periodically. Upstream and downstream, the job execution will take place as per the configuration.

After completing all the processes, you can see the visualisation shown in Figure 11. Green colour indicates the successful execution of a pipeline whereas red indicates an unsuccessful build. 

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